

Operational Amplifier

Definition: An Operational Amplifier, or **Op-amp** for short, is fundamentally a voltage amplifying device designed to be used with external feedback components such as resistors and capacitors between its output and input terminals.

Application:

- ☐ Mathematical operation (addition, subtraction, differentiation, integration etc.)
- ☐ Filtering (Band pass filter, low pass , high pass filter)
- ☐ Oscillator
- ☐ Zero crossing detector
- ☐ Function generator.

Why Operational amplifier is called operational amplifier ?

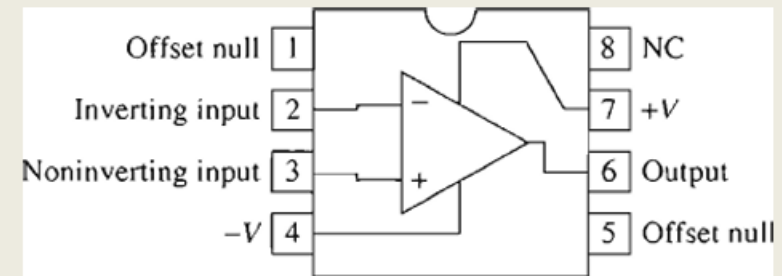
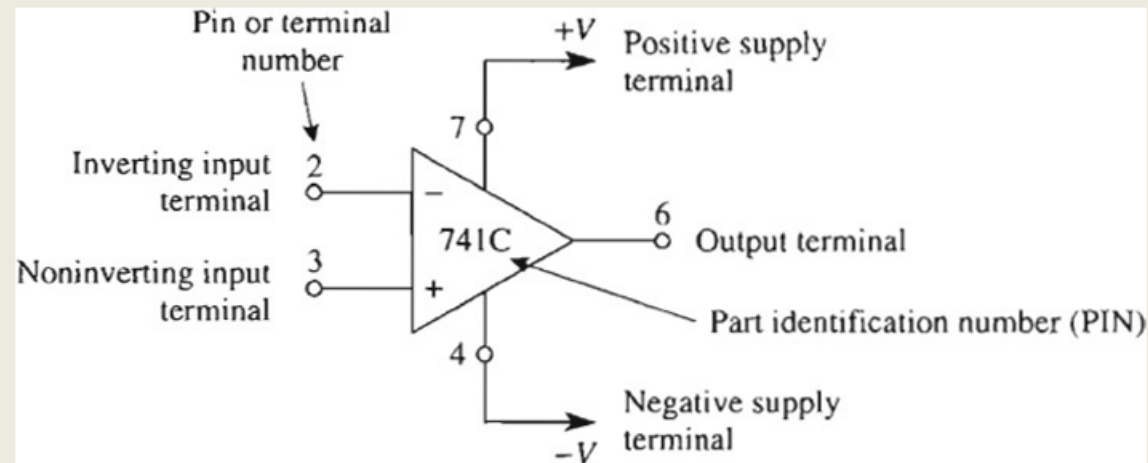
The name "operational amplifier" originates from the device's initial purpose in analog computers, where it performed mathematical "operations" like addition and integration. Its core function, signal "amplification," is also reflected in the name, highlighting its ability to boost electrical signals. Thus, "op-amp" concisely describes a component designed for both mathematical manipulation and signal amplification.



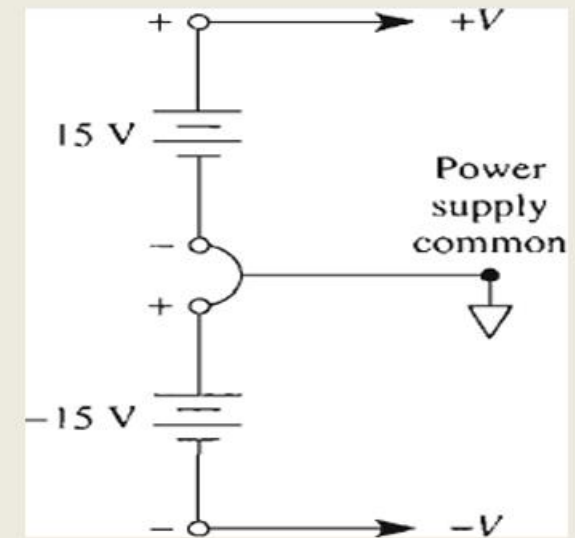
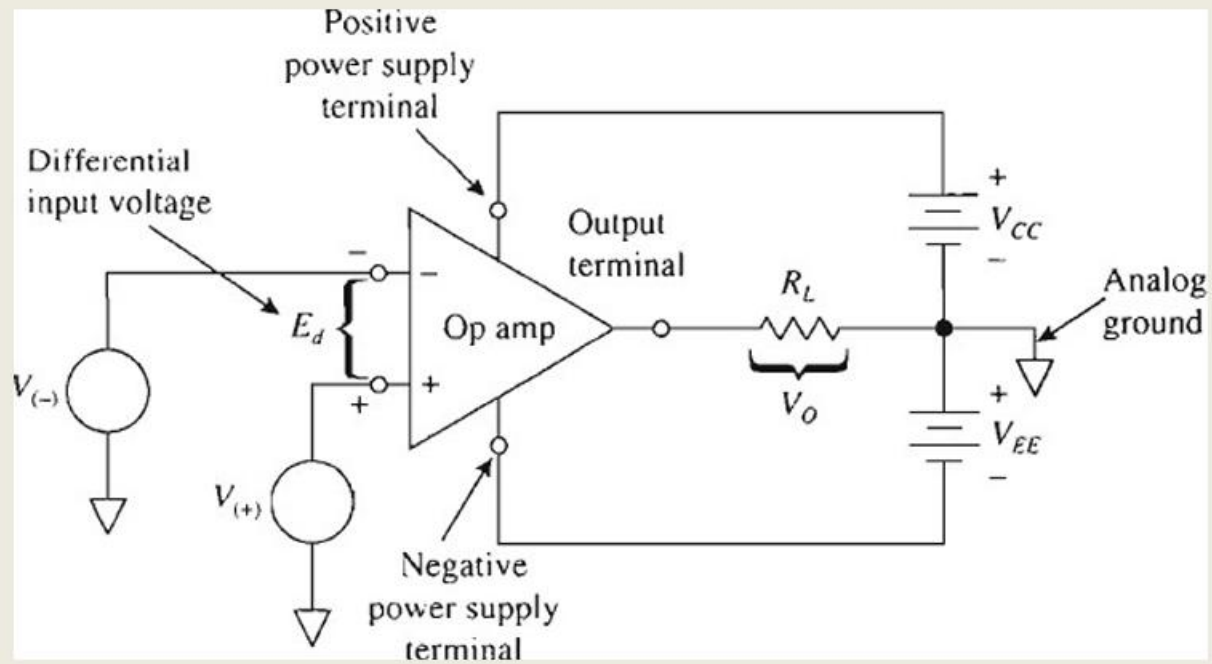
Fig 01: Practical Op-amp

Operational Amplifier (OP AMP)

An operational amplifier is an active circuit element designed to perform mathematical operations of addition, subtraction, multiplication, division, differentiation and integration.



Operational Amplifier (OP AMP)



Characteristics of Op-amp

Ideal Op-Amp Characteristics:

- Infinite open-loop gain
- Infinite input impedance
- Zero output impedance
- Infinite bandwidth
- Zero offset voltage
- Infinite CMRR (common-mode rejection ratio)
- Infinite slew rate
- Zero noise

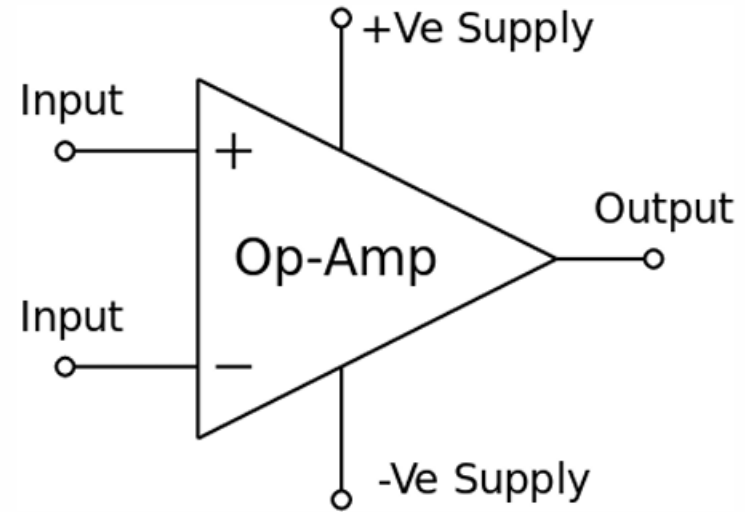
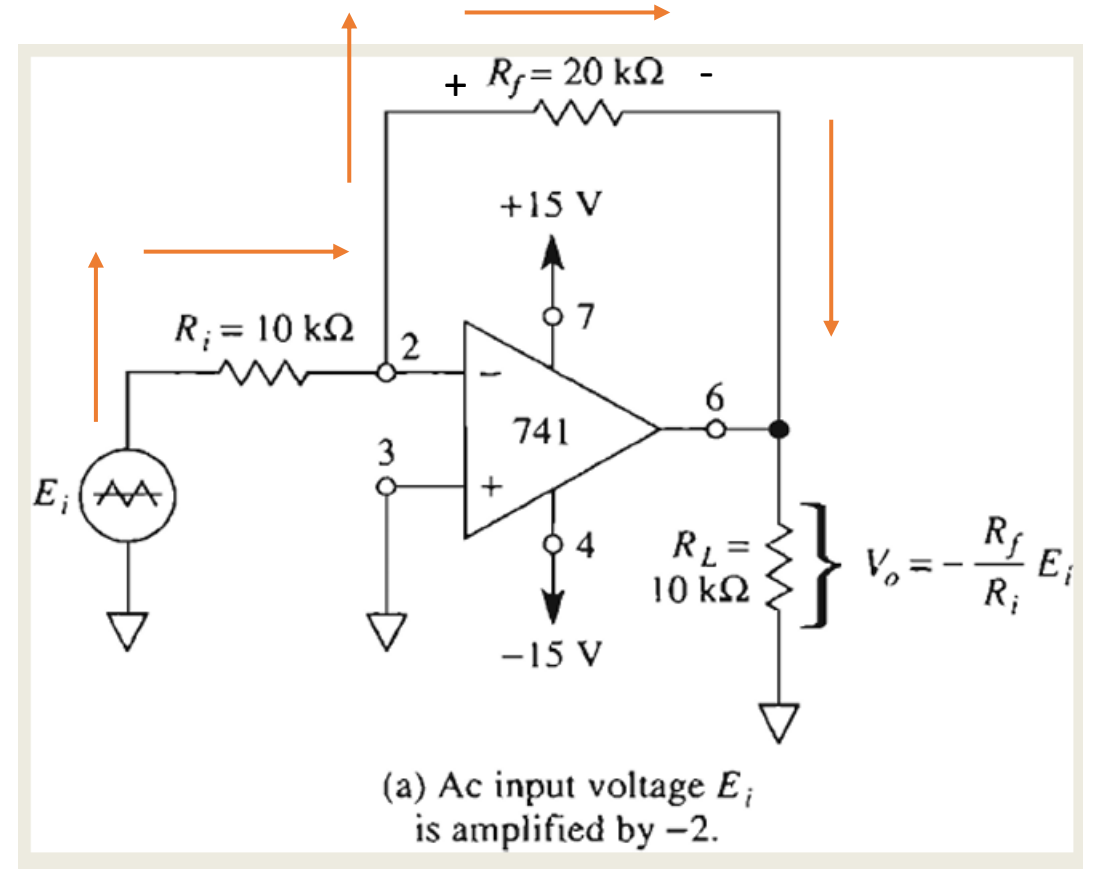
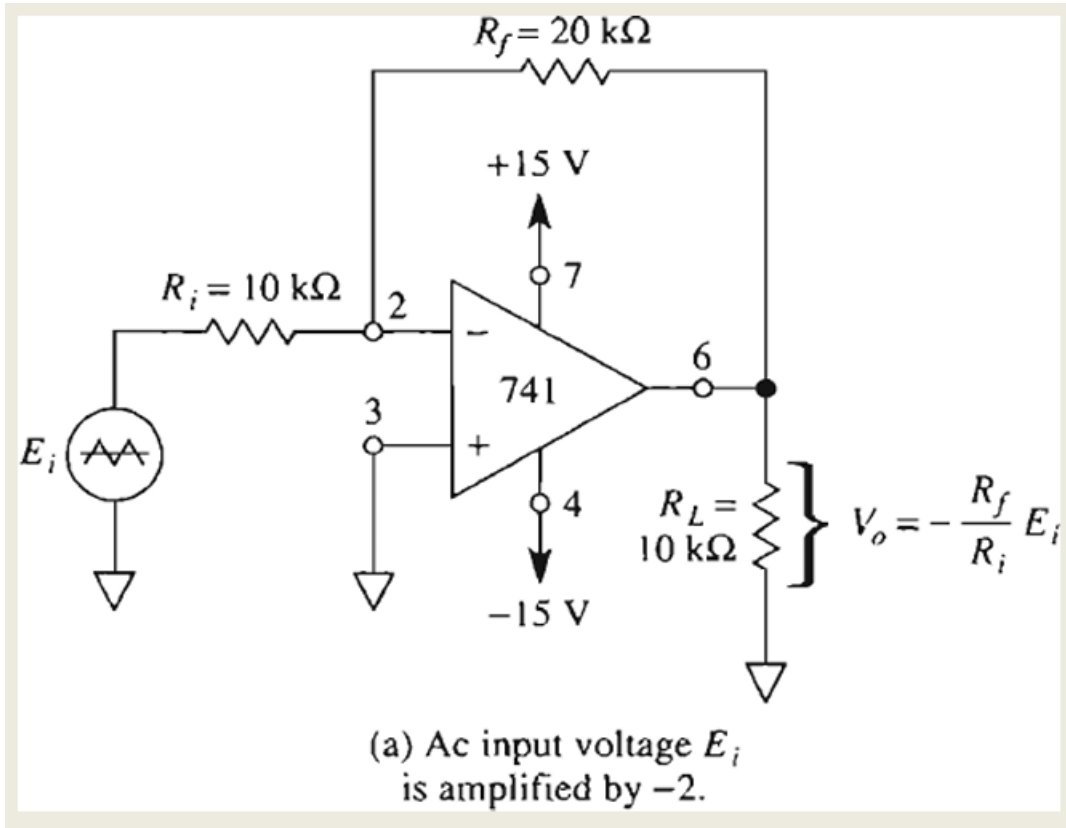


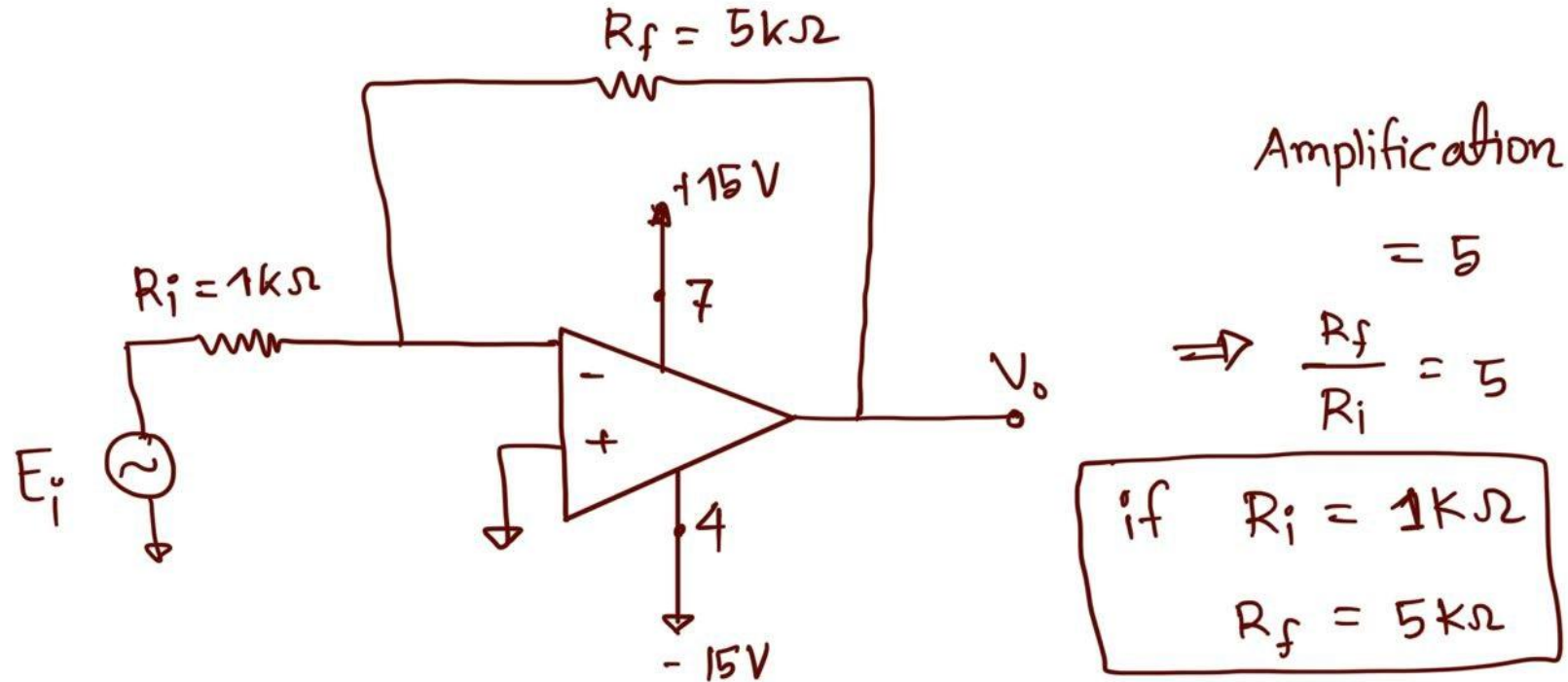
Fig 02: Symbol of Op-amp

Inverting Amplifier

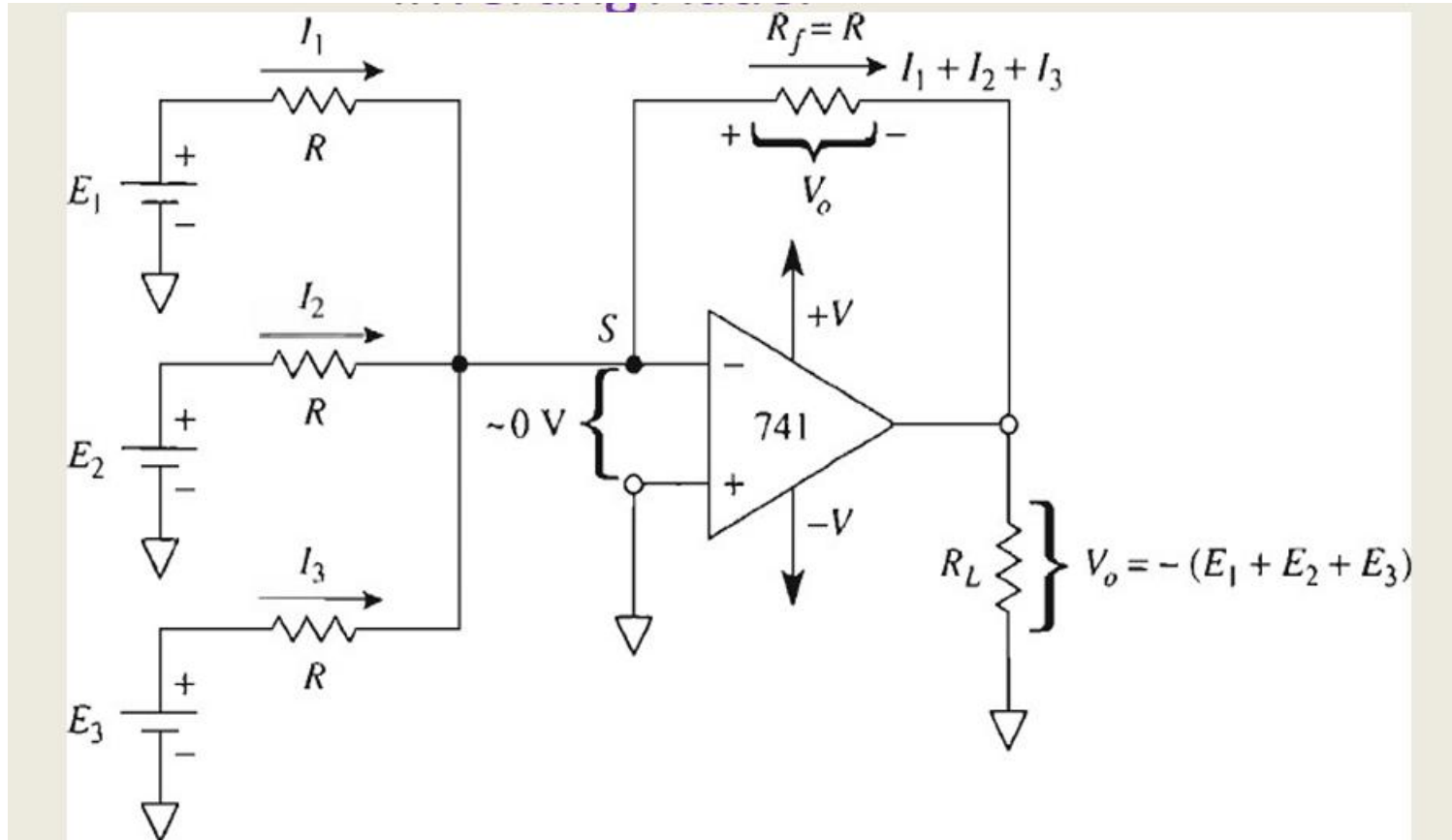


Inverting Amplifier

Question: Design an op-amp-based inverting amplifier that provides a voltage gain of 5.

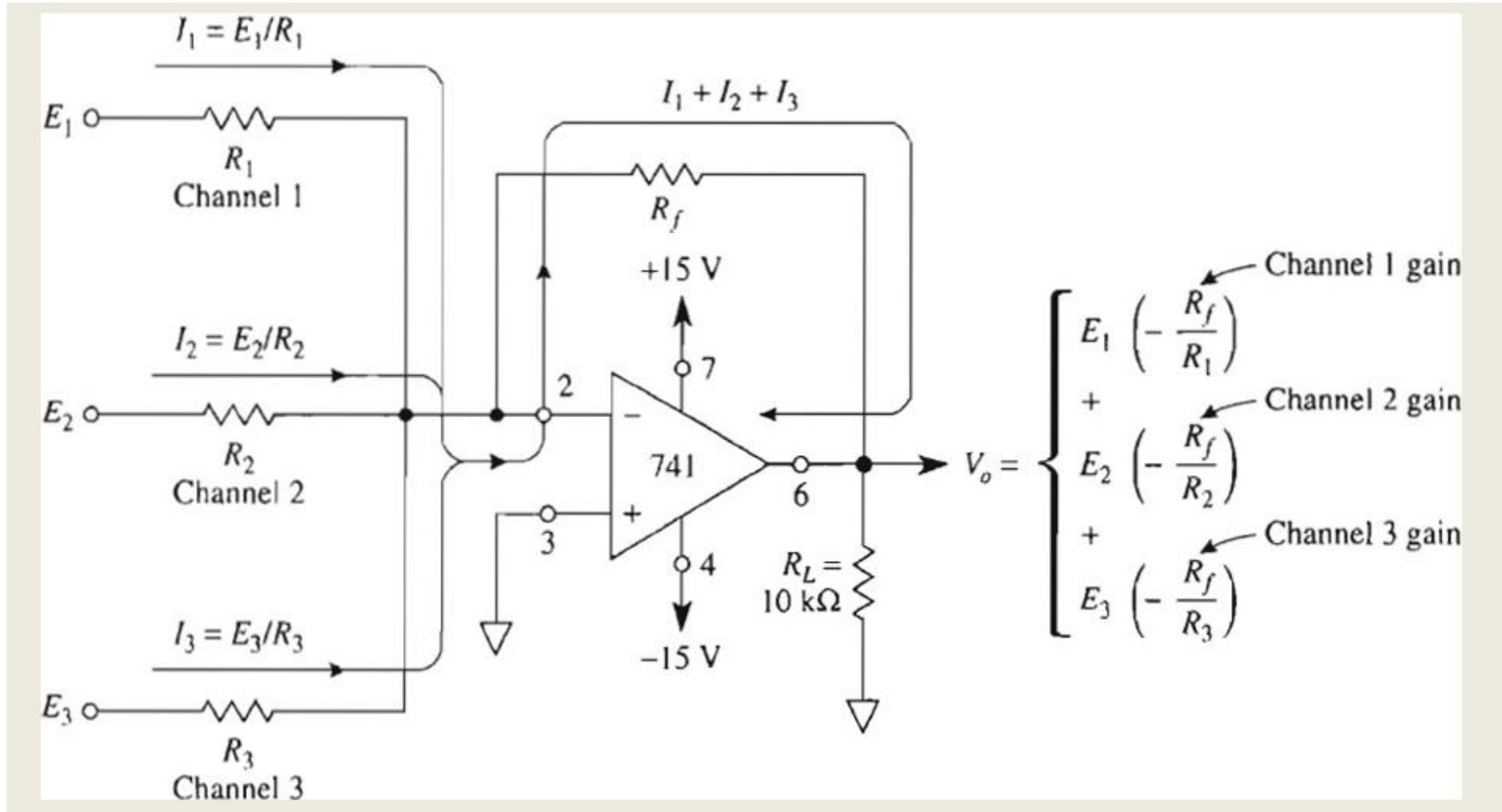


Inverting Adder

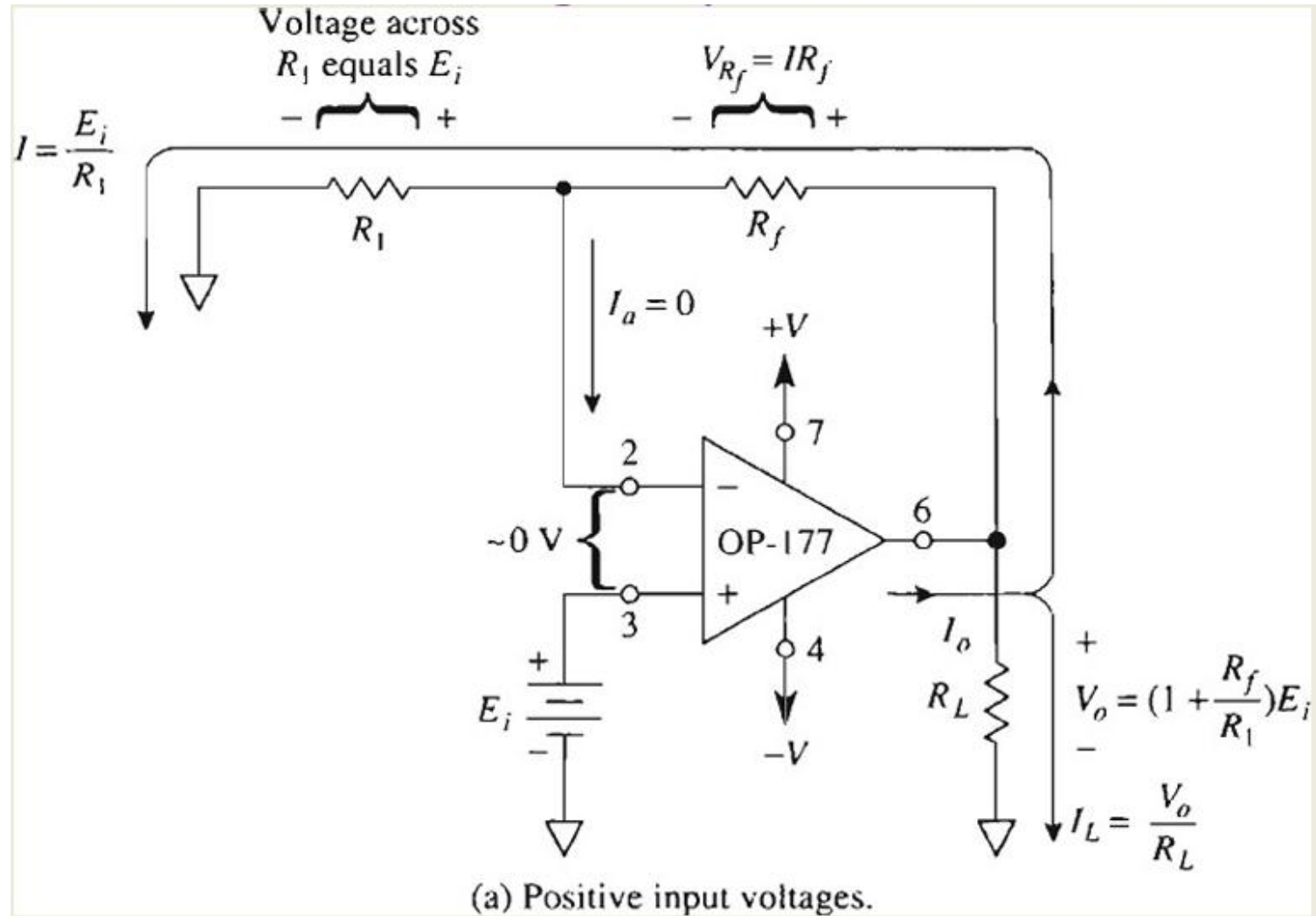


$$= -\frac{R}{R}E_1 - \frac{R}{R}E_2 - \frac{R}{R}E_3$$
$$= -(E_1 + E_2 + E_3)$$

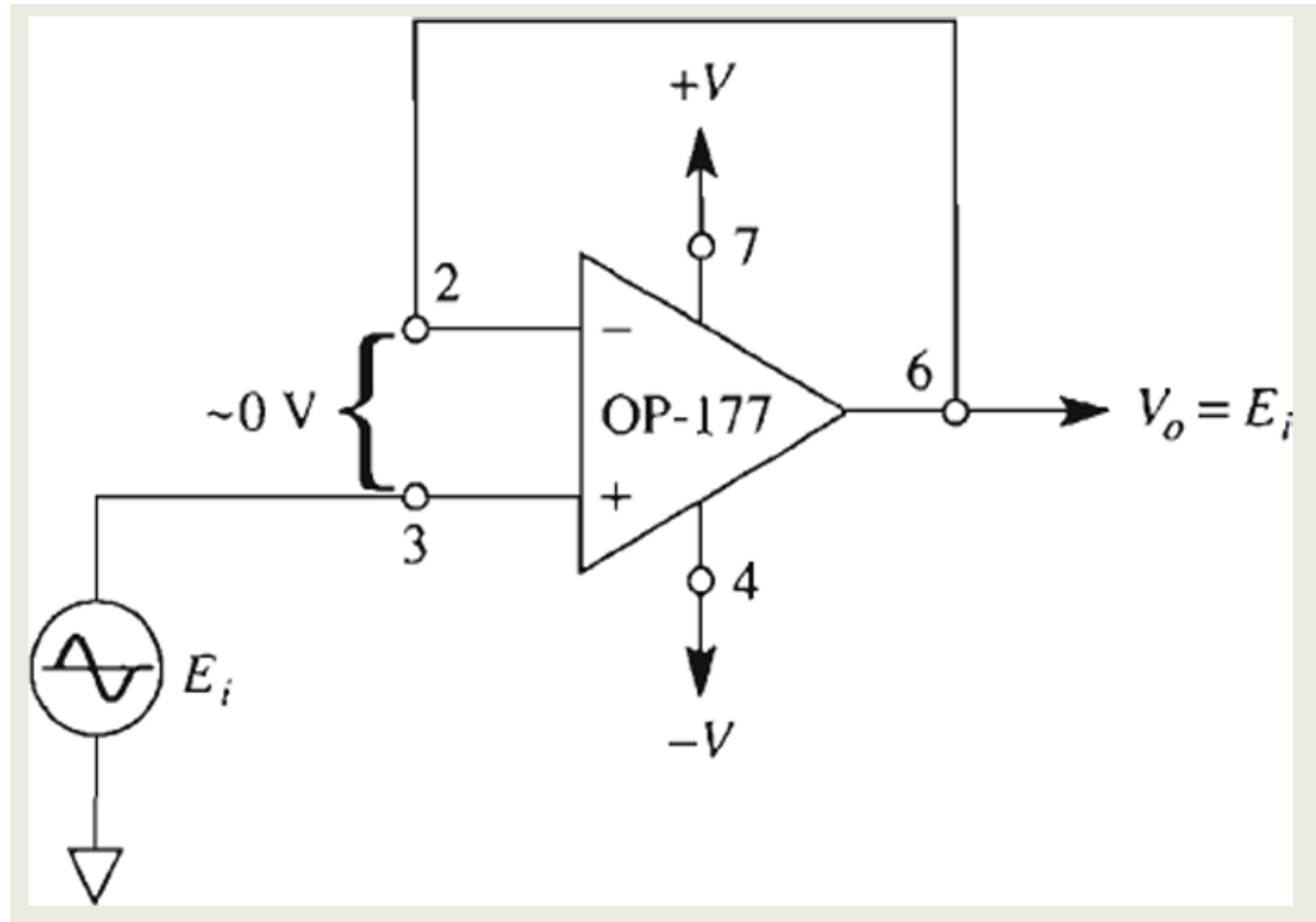
Multichannel Amplifier



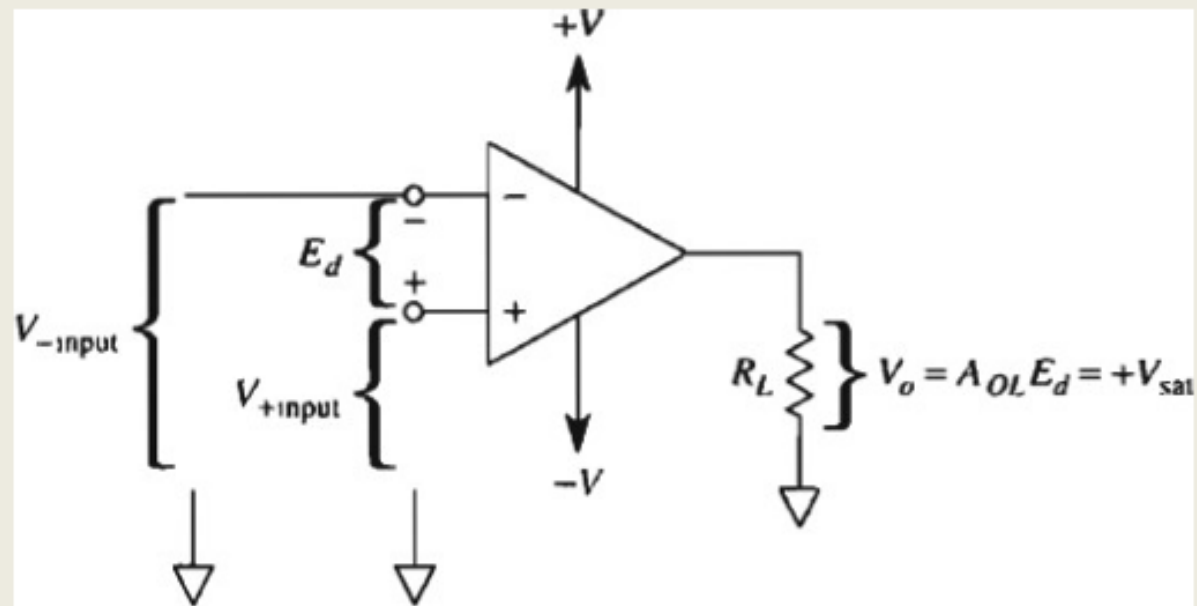
Noninverting Amplifier



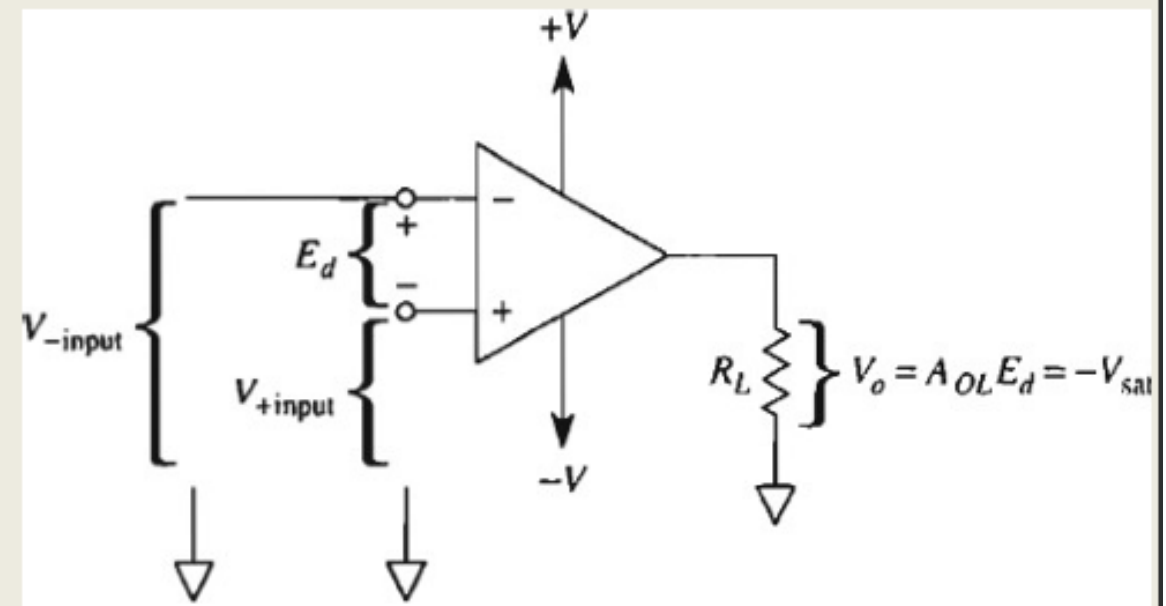
Voltage Follower



Ideal OP AMP Characteristics

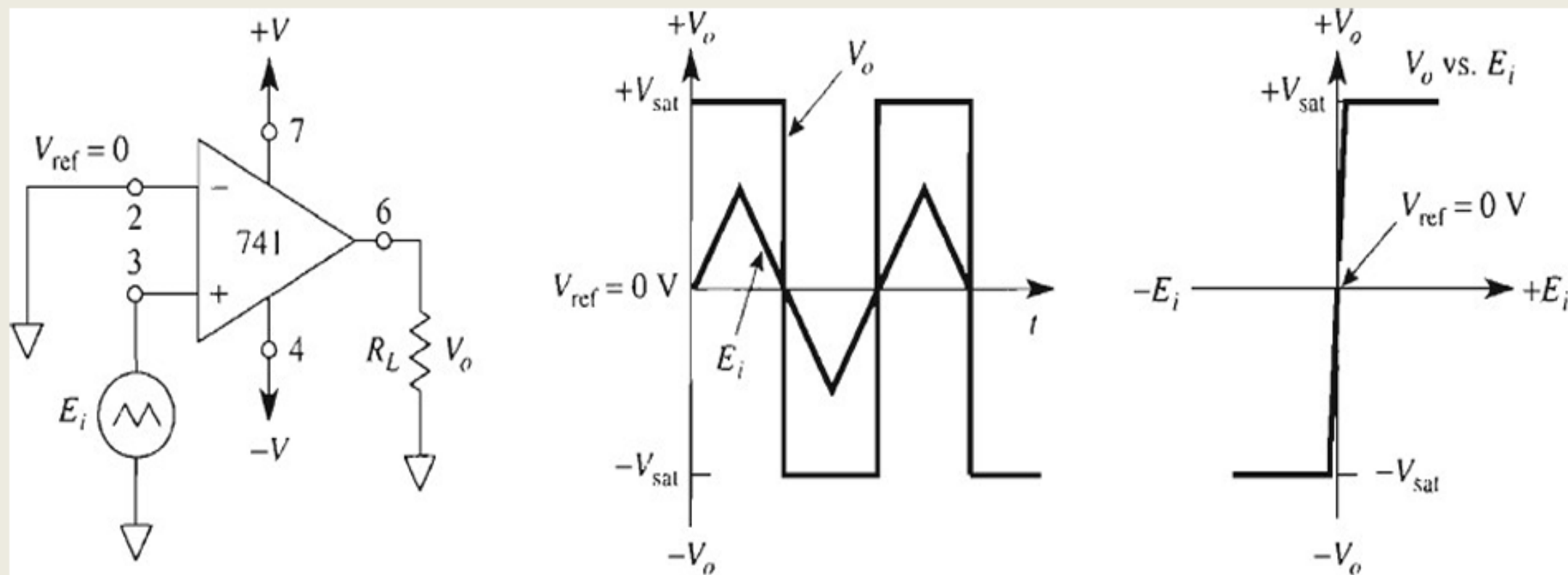


(a) V_o goes positive when the (+) input is more positive than (above) the (-) input, $E_d = (+)$.



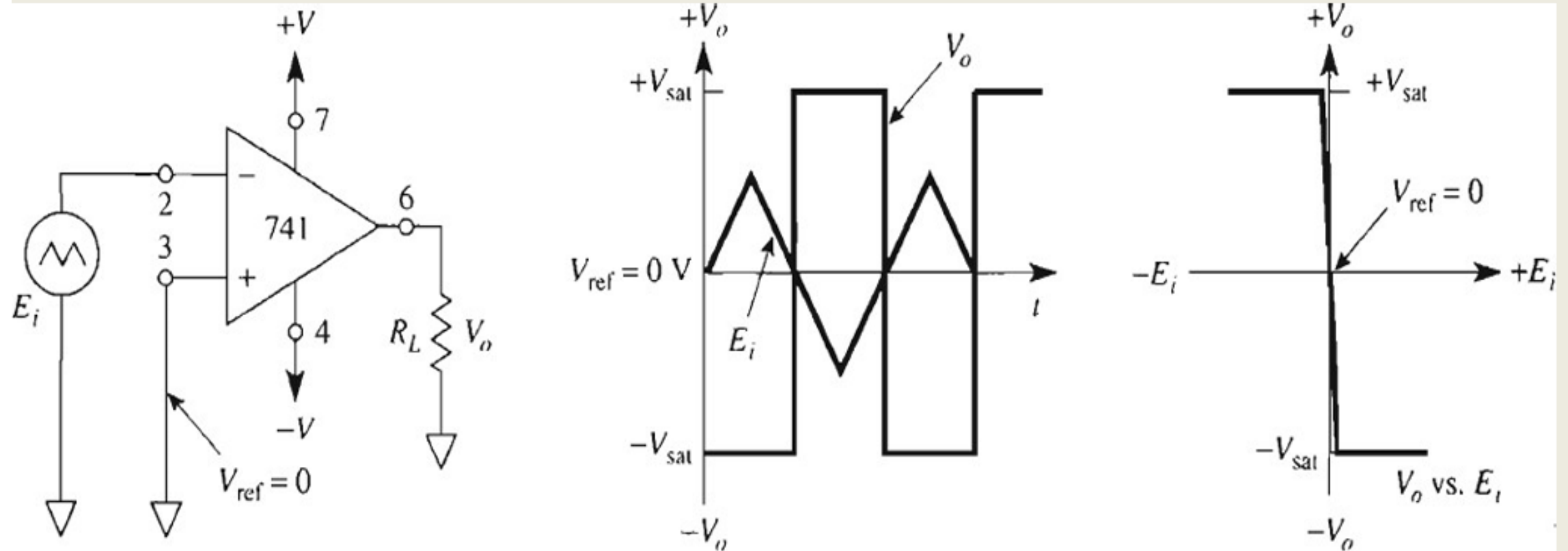
(b) V_o goes negative when the (+) input is less positive than (below) the (-) input, $E_d = (-)$.

Noninverting Zero-Crossing Detector



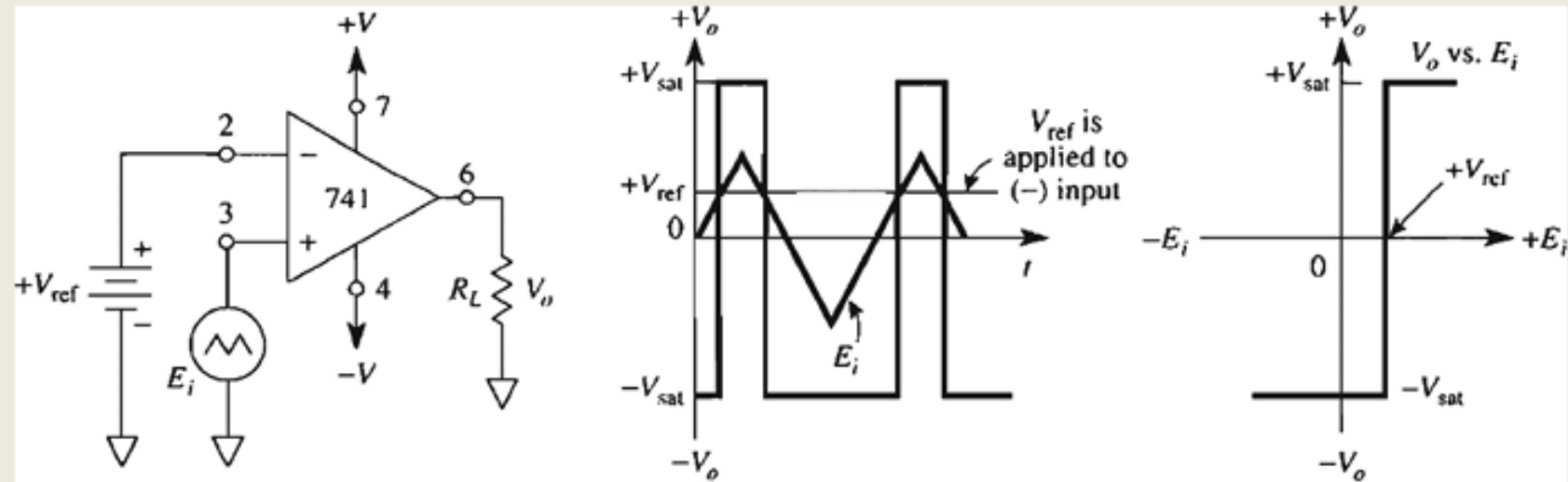
(a) Noninverting: When E_i is above V_{ref} , $V_o = +V_{sat}$.

Inverting Zero-Crossing Detector

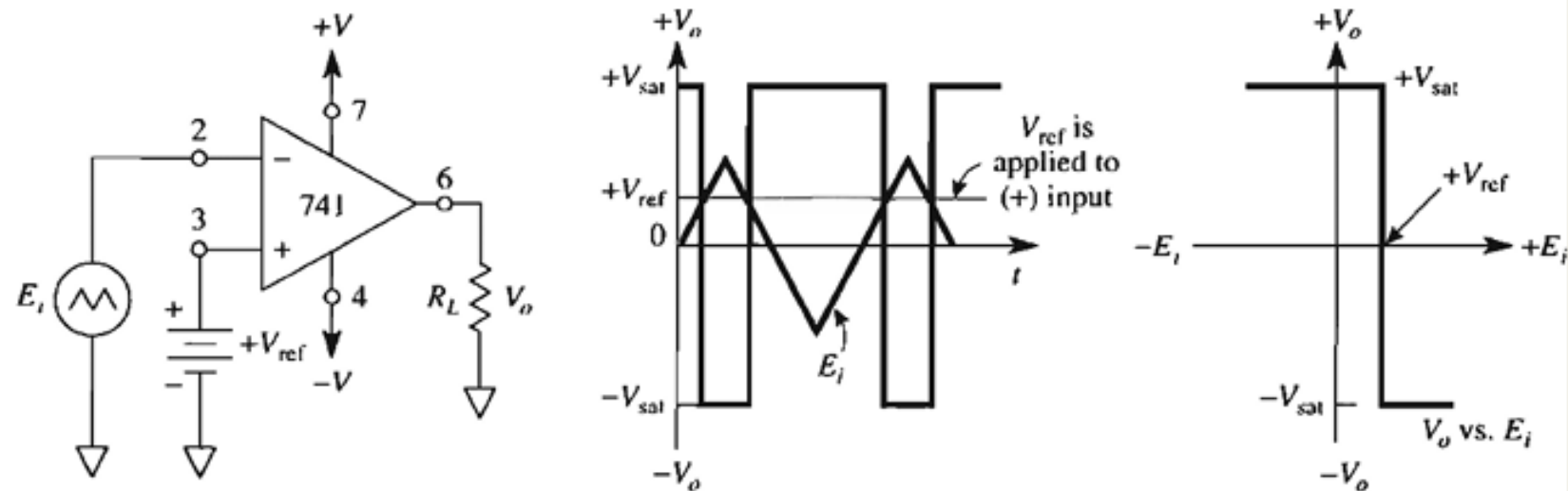


(b) Inverting: When E_i is above V_{ref} , $V_o = -V_{sat}$.

Positive Voltage-Level Detector

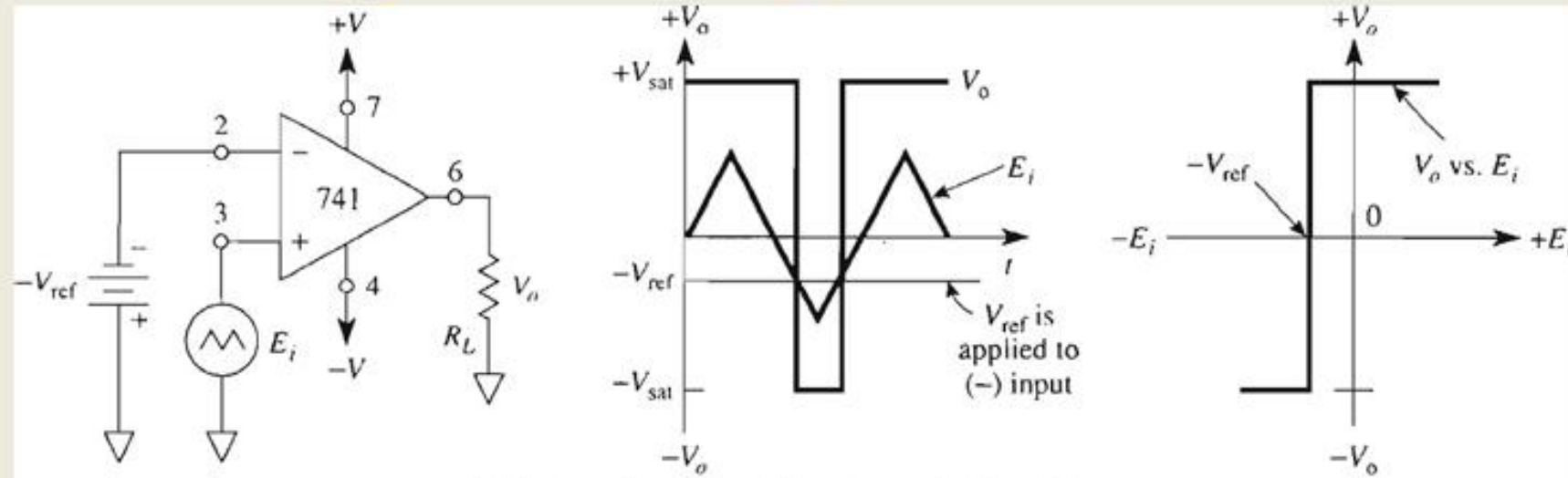


(a) Noninverting: When E_i is above V_{ref} , $V_o = +V_{sat}$.

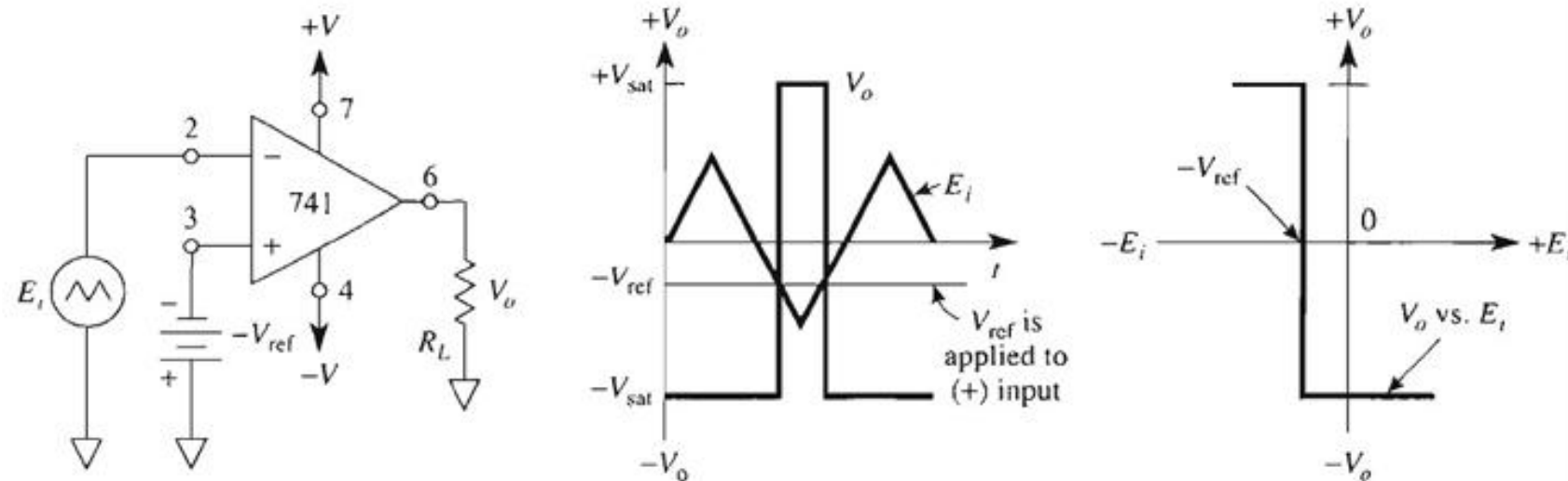


(b) Inverting: When E_i is above V_{ref} , $V_o = -V_{sat}$.

Negative Voltage-Level Detector



(a) Noninverting: When E_i is above V_{ref} , $V_o = +V_{sat}$.



(b) Inverting: When E_i is above V_{ref} , $V_o = -V_{sat}$.

Syllabus Covered:

1. BJT
2. JFET
3. MOSFET
4. Op-Amp

